

Extension: Chords of a circle



1 $x = 32$

2

a $x = 60^\circ$

b $x = 54^\circ$

c $x = 23$

d $x = 21$

e $x = 10$

f $x = 16$

3 $AB = 16$ cm

4 $x = 33^\circ$

5

a $OA = \sqrt{205}$

b $BP = \sqrt{530}$

6 $XN = 9$ cm

7 $YZ = 19$ cm

8 $SV = 26.25$ cm

9 $r = 17$ cm

10

To prove: $\triangle ABO \cong \triangle CDO$

Statements	Reasons
1. $AB = CD$	Given
2. $m\angle OEB = m\angle OFD = 90^\circ$	Given
3. $\overline{AO} \cong \overline{OB} \cong \overline{OD} \cong \overline{OC}$	Definition of radii
4. $\overline{AB} \cong \overline{CD}$	Definition of congruent segments
5. $\triangle ABO \cong \triangle CDO$	SSS congruence theorem

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To prove: $\triangle AOC \cong \triangle OBC$ 

Statements	Reasons
1. $\overline{OC} \perp \overline{AB}$	Given
2. $\overline{OA} \cong \overline{OB}$	Definition of radii
3. $m\angle ACO = \angle BCO = 90^\circ$	Definition of perpendicular segments
4. $\overline{OC} \cong \overline{OC}$	Reflexive property
5. $\triangle OAC \cong \triangle OBC$	RHS congruence theorem

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To prove: $\overline{AB} \cong \overline{DC}$

Statements	Reasons
1. $\angle AOB \cong \angle DOC$	Given
2. $m\angle AOB = m\angle DOC$	Definition of congruent angles
3. $\overline{OA} \cong \overline{OB} \cong \overline{OC} \cong \overline{OD}$	Definition of radii
4. $m\angle ACO = m\angle BCO = 90^\circ$	Definition of perpendicular segments
5. $\triangle AOB \cong \triangle DOC$	SAS congruence theorem
6. $\overline{AB} \cong \overline{DC}$	CPCTC

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To prove: $\triangle ABO \cong \triangle DEO$

Statements	Reasons
1. $\overline{AB} \parallel \overline{ED}$	Given
2. $\overline{OA} \cong \overline{OD}$	Definition of radii
3. $\angle BAO \cong \angle EDO$	If two lines are parallel, then the alternate interior angles are congruent (Alternate interior angles theorem)
4. $\angle AOB \cong \angle DOE$	Vertical angles theorem
5. $\triangle ABO \cong \triangle DEO$	ASA congruence theorem



Additional questions

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a $x = 4$

b $m\widehat{RS} = 40^\circ$

c $m\widehat{RQ} = 88^\circ$

d $x = 3$

15 $m\widehat{BC} = 118^\circ$

16

a True

b False

c False

d False

17 30 feet

18

a $y = 7$

b $UV = 11$ cm

19

a $x = 16$

b $m\angle SPT = 160^\circ$

20

a $x = 7$

b $MN = 62$

21

a Since chords $\overline{AD}$ and $\overline{EB}$ are congruent, their corresponding central angles are congruent.

b $m\widehat{AD} = 122^\circ$

c $m\widehat{DB} = 76^\circ$

22 The segment length of eyebrow which cuts through the left eye is equal in length to that which cuts through the right eye.

23

a $x = 6$

b $m\angle 1 = 49.5^\circ$

c $m\widehat{EDC} = 125^\circ$

d $m\angle 2 = 61.5^\circ$

24

To prove: $\widehat{AB} \cong \widehat{DC}$

Statements	Reasons
1. $\angle AOB \cong \angle DOC$	Given
2. $\overline{AO}, \overline{OC}, \overline{OD},$ and $\overline{OB}$ are radii of circle O	Definition of radii
3. $\overline{AO} \cong \overline{OB} \cong \overline{OD} \cong \overline{OC}$	Definition of radii
4. $\triangle AOB \cong \triangle DOC$	SAS congruence theorem
5. $\overline{AE} \cong \overline{BD}$	CPCTC
6. $\widehat{AB} \cong \widehat{DC}$	If two chords in a circle are congruent, then their corresponding arcs are congruent

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To prove: $\overline{JP}$ bisects $\widehat{KJM}$

Statements	Reasons
1. circle P	Given
2. $\overline{KM} \perp \overline{JP}$	Given
3. $\overline{KP} \cong \overline{PM}$	Definition of radii
4. $\angle PLK \cong \angle PLM$	Definition of right angles
5. $\overline{PL} \cong \overline{PL}$	Reflexive property
6. $\triangle PLK \cong \triangle PLM$	SAS congruence theorem
7. $\angle MPJ \cong \angle KPJ$	CPCTC
8. $\widehat{KJ} \cong \widehat{JM}$	If two central angles are congruent, then their corresponding arcs are congruent
9. $\overline{JP}$ bisects $\widehat{KJM}$	Definition of bisector

26 Since $\widehat{BC} \cong \widehat{ED}$, we know that $\overline{BC} \cong \overline{ED}$. Since $\angle CBE$ and $\angle EDC$ are inscribed in the same arc, they are congruent. Since $\angle BCE$ and $\angle DCE$ are inscribed in congruent arcs, they are also congruent. This means by AAS, $\triangle BCE$ and $\triangle DEC$ are congruent, so $\overline{CD} \cong \overline{EB}$ by CPCTC